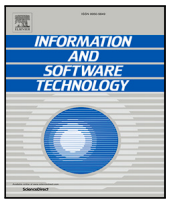
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MLOps best practices, challenges and maturity models: A systematic literature review

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ABSTRACT

Context: Agile and DevOps methodologies have revolutionized software development, leading to increased efficiency and reliability in product delivery. Building on this success, Machine Learning Operations (MLOps) has emerged to streamline the development and deployment of machine learning (ML) models, addressing challenges unique to ML workflows.

Objectives: This study aims to explore the complexities organizations face when adopting MLOps, focusing on three main challenges: the lack of standardized practices, difficulties in maintaining model consistency and scalability, and ambiguities in assessing MLOps maturity. The study also aims to identify best practices and common pitfalls, contributing to a clearer understanding and standardization of MLOps.

Methods: A comprehensive literature review was conducted, analyzing 45 articles that address MLOps best practices, challenges, maturity models, and lessons from previous implementations. This review categorizes findings to provide insights into successful and unsuccessful MLOps applications.

Results: The study identifies nine best practices, eight common challenges, and five maturity models relevant to MLOps adoption. Key lessons from successful and unsuccessful MLOps implementations are outlined, with a focus on improving standardization and reducing ambiguity in MLOps practices.

Conclusion: The findings highlight the importance of establishing standardized MLOps practices to address the unique challenges of machine learning workflows. By categorizing best practices, maturity models, and lessons learned, this study aims to contribute to a robust MLOps framework that enhances the reliability and scalability of machine learning in production environments.

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1. Introduction

Artificial intelligence (AI), driven by advancements in machine learning (ML), is revolutionizing various sectors, from healthcare to finance. However, successfully deploying and maintaining AI models in production presents significant challenges. This necessitates a robust and efficient development approach, where MLOps comes into play. MLOps, a combination of ML and Operations, bridges the gap between the development and operations aspects of the ML life cycle, enabling organizations to streamline the entire process from model development to deployment and maintenance [1–4].

MLOps encompasses a set of best practices, tools, and automation techniques that accelerate model delivery, improve model reliability, and foster collaboration among data scientists, ML engineers, and IT teams [1,2,5]. By automating key stages of the ML life cycle, such as data preparation, model training, deployment, and monitoring, MLOps:

1. Reduces friction and accelerates model delivery: Automated pipelines streamline the ML life cycle, leading to faster deployment and experimentation [6–8].
2. Improves model reliability and performance: continuous monitoring and feedback loops ensure models stay efficient and adapt to changing data [1,7,9].
3. Increases collaboration and communication: It fosters closer cooperation between data scientists, ML engineers, and IT teams [2,10].
4. Reduces costs and technical debt: Automated workflows minimize human error and simplify model maintenance [2,3].

The importance of MLOps is underscored by its rapid adoption across industries. According to [2], 70% of organizations are already investing in MLOps tools and platforms, reflecting the growing recognition of its value in achieving successful AI initiatives. However, challenges remain, with 55% companies citing the lack of adequate MLOps practices as a major obstacle to the deployment of ML models [1,2].

Hence, this research aims to contribute to a deeper understanding of the MLOps landscape and provide insights for organizations seeking to effectively leverage AI in their operations. By systematically reviewing the existing literature on MLOps best practices, challenges, and maturity models, our objective is to understand the challenges faced by organizations and identify effective best practices to address them.

2. Context

DevOps has become a critical practice for modern software development, enabling organizations to accelerate development cycles, increase deployment frequency, and improve software quality [1,2]. By bridging the gap between the development and operations teams, DevOps promotes collaboration, automation, and continuous improvement [11]. Key concepts within DevOps include Continuous Integration (CI) and Continuous Delivery (CD). CI emphasizes the frequent integration of code into a shared repository, enabling early detection and resolution of integration issues [12,13]. CD focuses on automating the release process, ensuring that the software is always in a deployable state [3,14]. Fig. 1 illustrates how the CI/CD pipeline intersects with the development and operation processes [15].

In the context of Machine Learning (ML), MLOps extends the principles of DevOps to the entire ML life cycle where MLOps encompasses a set of practices, tools, and technologies that facilitate the development, deployment, and maintenance of ML models in production. It involves a collaborative approach between data scientists, software engineers, and operations teams [1,16,17].

MLOps introduces additional concepts, such as Continuous Training (CT) and Continuous Monitoring (CM). CT enables automatic retraining and redeployment of models to address issues such as data drift [18], ensuring that the models remain accurate and effective over time [3,19]. CM involves continuous monitoring of model performance in production, allowing early detection of anomalies and proactive intervention [20].

To achieve these objectives, MLOps leverages automation throughout the ML life cycle, including:

1. Data engineering: Data collection, cleaning, transformation, and feature engineering.
2. Model training and evaluation: Training, validation, and evaluation of ML models.
3. Model deployment: Deployment of models to production environments.
4. Model monitoring: Tracking model performance, detecting anomalies, and triggering retraining.

By implementing MLOps practices, organizations can streamline the ML life cycle, accelerate time-to-market for new ML applications, and improve the overall quality and reliability of their ML systems. MLOps concept is increasingly vital for organizations aiming to streamline ML model development and deployment. However, MLOps adoption is challenged by a lack of standardized practices, difficulties in maintaining model consistency, and unclear maturity models. These issues result in inefficiencies and increase the risk of project failures.

This research addresses these gaps by conducting a systematic literature review (SLR) to identify best practices, challenges, maturity models, and lessons learned in MLOps. By analyzing 45 academic articles, the study aims to provide clear guidelines and actionable insights to help organizations effectively implement MLOps, enhancing model management and reducing time-to-market.

3. Research methodology

This research delves into the practical complexities of MLOps by conducting a systematic literature review (SLR) by following the procedures of Kitchenham [21] to carry out this SLR. We aim to unearth the challenges, best practices, maturity models, and invaluable lessons learned from diverse organizations navigating the journey of implementing MLOps. To achieve this, we will explore various academic publications, utilizing established databases and employing a targeted keyword search. Through a rigorous screening process, we will curate a collection of relevant studies, meticulously extracting and analyzing their insights. By synthesizing these findings, we will paint a comprehensive picture of the MLOps landscape, revealing common pitfalls, successful strategies, and crucial lessons for navigating this dynamic field. Ultimately, this research aims to equip practitioners with actionable insights and practical solutions to navigate the intricacies of MLOps implementation, paving the way for smoother, more efficient transitions from code to impactful ML models.

The research methodology can be divided into four main stages with each having a different number of steps as follows in Fig. 2.

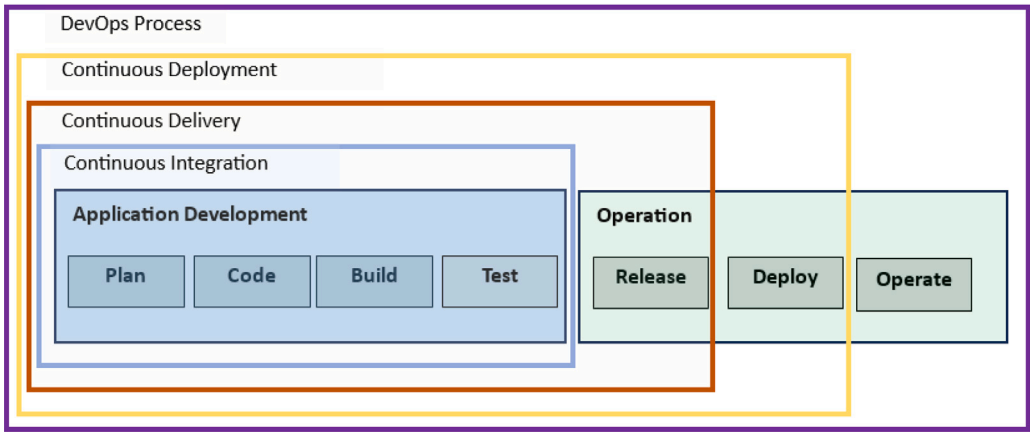


Fig. 1. DevOps CI/CD pipeline intersection with development and operation processes [15].

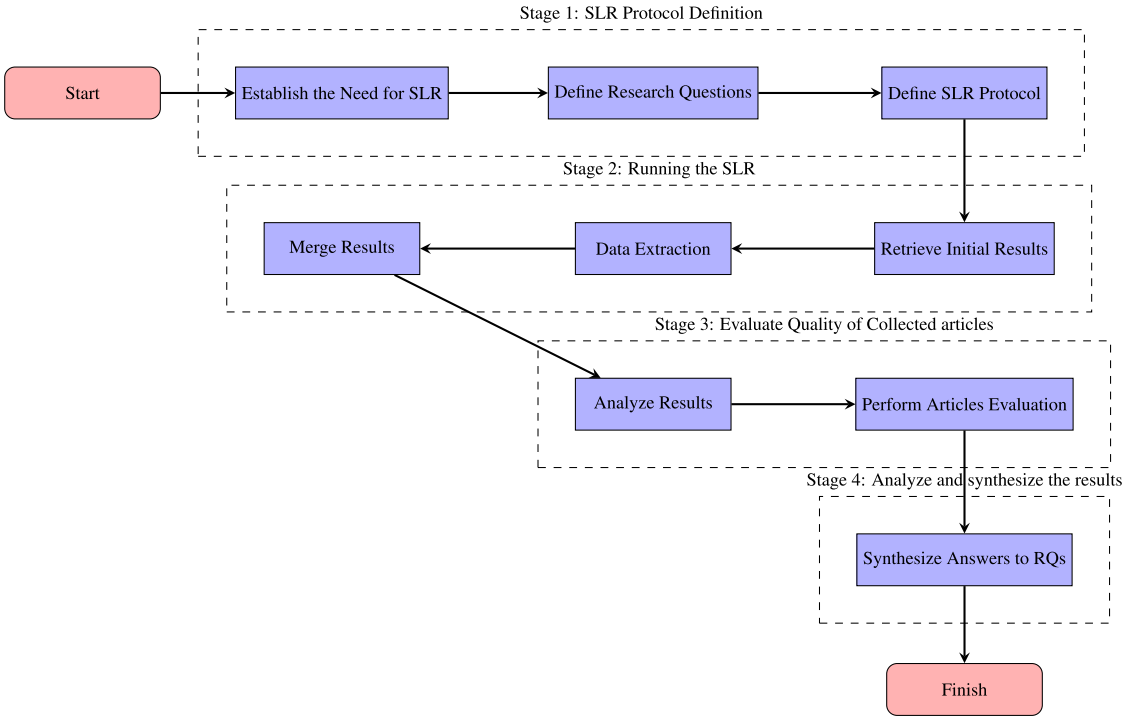


Fig. 2. The phases and steps of the methodology.

The first stage, the SLR Protocol Definition, contains three steps; The first step was implicitly addressed as it suggested that there was a growing body of literature on MLOps practices, challenges, and maturity models, indicating a need for an SLR. The second step of the first phase was to define research questions that are focused on identifying MLOps' best practices, challenges, maturity models, and lessons learned from successful and unsuccessful implementation. These questions arise from the need to understand and categorize various aspects of MLOps based on existing literature. The stage of Defining the SLR Protocol involved specifying criteria for selecting relevant articles and conducting searches across databases. It is worth noting that the output of the first phase is the SLR protocol. The second stage, Running the SLR, includes determining criteria for the inclusion and exclusion of the articles, search strategies, and data extraction methods. The step of retrieving initial results determines 45 articles to be examined. In the step of Data Extraction, data was systematically extracted from the included articles, focusing on key findings, best practices, challenges, maturity models, and lessons learned related to MLOps implementation. In the step of Merge Results, the extracted data from individual articles are organized and consolidated to facilitate analysis. This involves categorizing articles, identifying common themes, and preparing for further analysis. The output of the second phase was collected articles. The third stage, Evaluating the Quality of Collected articles, evaluates how effective and reliable the chosen articles were in addressing the RQs by applying quality criteria, and it is worth mentioning that this stage is performed after the analysis of the articles to be able to determine how relatable and beneficial they were within the scope of the research. The final stage includes analyzing results which identifies patterns, trends, and insights relevant to the research questions. This analysis aims to synthesize the findings from the SLR and draw meaningful conclusions. This involves considering the strengths and limitations of each study and evaluating their implications for the overall findings of the review. Finally, the findings from the SLR are synthesized to address the research questions and objectives. This involves summarizing key findings, categorizing best practices and challenges, and drawing conclusions based on the evidence gathered.

Table 1
Result by search expression.

Search String	ACM DL	IEEE Xplore	Science Direct	Springer Link	Google Scholar	Total
"MLOps"	153	95	90	86	34	458
"machine learning life cycle"	40	186	13	258	12	509
"machine learning operations"	102	95	103	163	44	507
"machine learning" AND "deployment pipeline"	52	150	35	6	8	251
"machine learning" AND "MLOps practices"	20	26	5	25	11	87
"machine learning" AND "deployment challenges"	68	93	78	112	18	369
"machine learning" AND "maturity model" AND "deployment"	57	3	166	34	6	266
"machine learning" AND "continuous delivery"	164	44	128	36	20	392
"machine learning" AND "continuous integration"	591	167	373	90	17	1238
Total	1247	859	991	810	170	4077

3.1. Research questions

To gather preliminary data for this SLR, a Google Scholar search was conducted with the terms "MLOps" and "machine learning operations" to gather publications on the subject. The following research questions (RQs) were formulated with the assistance of the information gathered, which is related to the overall goal of the study, which is to identify the studies that address MLOps best practices, challenges, maturity models, and successful and unsuccessful implementation of MLOps. While the previously mentioned aspects of MLOps played a vital role in defining the RQs, those aspects were found to contain many research gaps, and that is why we focused on them. Those RQs are attempting to address as many best practices as possible which helps get a clearer idea about MLOps, as many challenges as possible to then be able to mitigate and curate them, all available MLOps maturity models which helps understand MLOps maturity, and the lessons learned from successful and unsuccessful MLOps implementation.

1. RQ1 - What challenges are commonly encountered in the implementation of MLOps in various organizations?
2. RQ2 - What are the practices in MLOps for developing, deploying, and maintaining machine learning models?
3. RQ3 - What maturity models or frameworks have been proposed for assessing an organization's level of MLOps maturity, and how are they structured?
4. RQ4 - What are the key lessons learned from successful and unsuccessful MLOps implementations?

3.2. Search strategy

Searching electronic databases was the main approach used for this SLR. These included Springer Link, IEEE Xplore, Science Direct, and the ACM Digital Library. These databases were chosen based on several criteria, such as the extent to which they covered the subject, the frequency of their updates, the accessibility of all study content, the effectiveness of the automatic search engine, the ability to export results or integrate with other extensions to enable auto search playback and export. After collecting the needed articles, Google Scholar was also checked to add more relevant articles from other repositories. Since Google Scholar returns results from other databases, which we included in our search, such as ACM DL, IEEE, we considered removing duplicate publications directly from the returned results to eliminate unnecessary or duplicated articles. The start date of this study was January 1, 2019, considering all available studies until January 31, 2024.

3.3. Search terms

Key terms and expressions that were later employed as search expressions were found from the preliminary search that was conducted to contextualize the research. The search strings shown in Table 1 were created using the terms that were chosen. Some trial searches were conducted using these terms to determine the best combination of phrases to establish the search expressions that would be used in this systematic research. After evaluation, it was determined that the most effective approach would be to do a series of separate searches and then compile the articles that were discovered, even at the cost of utilizing a more complex search phrase that would limit the studies that deal with the research topic. The search was done using all fields including title, abstract, primary keywords, and article content to increase the number of articles that may be chosen. After applying the defined search terms to the electronic search bases, the search engines generated 4077 results. Table 1 displays the search terms that were employed as well as the full results from the electronic search database.

3.4. Selection criteria

The inclusion and exclusion criteria used to choose and eliminate the studies to be examined in the systematic review, taking into account the objectives of the research and scope, are explained in Table 2. Following the establishment of the selection criteria, they applied to the studies that were found using electronic search databases. As seen in the bottom right corner, the total number of articles was 4077. After removing 522 duplicate articles, 3355 articles remained to be evaluated. The titles of the articles were then checked against the specified inclusion and exclusion criteria. 629 articles were selected directly and 2926 articles were left to be excluded. Next, after reading the abstract of the included articles, 166 were chosen after applying the exclusion and inclusion criteria to them again, which excluded 463 articles. Finally, after reading the entire text, 121 of the remaining articles were excluded and 45 articles were examined and analyzed. Fig. 3 details the process described in this section for selecting the studies. For the sake of clarity, Table 3 shows how many articles were retrieved from each database.

3.5. Quality assessment

Evaluation of the articles' quality was necessary, given that the chosen studies represent the widest range of research types. To detect potential biases in the study's findings, a set of parameters that are examined in each investigation are correlated with the quality standards. Table 4 presents the quality criteria that were utilized in this research. The following scores were given to the articles based on

Table 2
Inclusion and exclusion criteria.

Inclusion criteria (IC)	Exclusion criteria (EC)
IC1 Studies that address MLOps in general.	EC1 Publication not in English.
IC2 Published between 2019 and Jan. 2024.	EC2 Studies that do not allow access to its content.
IC3 Studies that focus on practices, challenges, lessons learned and maturity models of MLOps.	EC3 The study was not related to MLOps.
IC4 Manuscripts closely aligned with the topic.	EC4 articles that do not address research questions.
IC5 The existence of the terms "MLOps" or "Machine Learning Operations" or "Machine Learning" in the title of the article.	EC5 Studies with no full text available.

Table 3
The number of articles per database after applying the inclusion/exclusion criteria.

Search engine	ACM	IEEE	Direct Science	Springer Link	Google Scholar
Number of articles retrieved	4	15	3	5	18

Table 4
Quality criteria based on [22].

QC	Criteria
1	The study reports a clear discovery based on evidence and argument
2	The study presents a research project and not an expert opinion
3	The number of research questions that are tightly coupled with the manuscript
4	The objective of the study was clearly defined
5	The study fully described the context analyzed

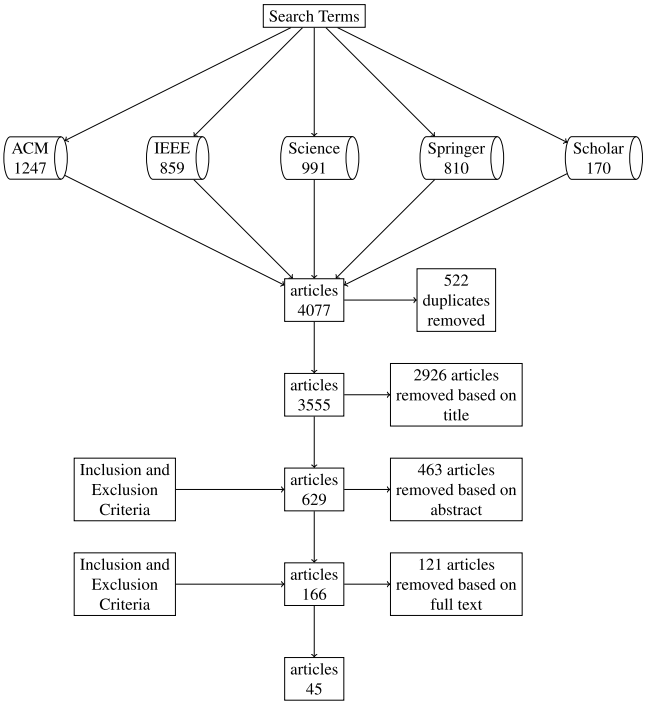


Fig. 3. Study selection protocol.

their evaluation under each criterion: If the response to the criteria is "Yes", the article receives one point; if it is "No", it receives zero. Based on the criteria for quality assessment, an article can receive up to a maximum of five points. Despite our quality assessment of these articles, we believe that publishing these articles in the credible journals and publications venues that we have selected to conduct this SLR maintains the level of quality of these publications. Table A.9 presents the findings from the assessment of the chosen articles in relation to the quality criteria outlined in this SLR and it is placed in the Appendix.

4. Data analysis

Data analysis involved a two-step process. First, we extracted key information from each selected study, including MLOps practices, challenges, lessons learned, and maturity model details. Second, We synthesized the findings across studies to identify key challenges, best practices, maturity models in MLOps, and lessons learned, ultimately providing answers to the specified research questions.

4.1. RQ1 - What challenges are commonly encountered in the implementation of MLOps in various organizations?

This question underscores the complexities and hurdles often faced when integrating MLOps practices into organizational workflows. By examining these challenges, we aim to identify common pitfalls and obstacles that organizations encounter during MLOps implementation. Those challenges include:

1. Lack of Necessary Skills and Culture: the successful implementation of MLOps requires a combination of cross-functional skills and a collaborative culture, which are often lacking within organizations. MLOps projects require expertise across software engineering, data science, and operations, making it challenging to find the right talent with the necessary skill set [23]. This challenge is compounded by the need for effective collaboration across different teams, including developers, data scientists, and IT personnel [8]. Moreover, managing the complexity of ML systems and workflows demands not only technical proficiency but also cultural shifts within organizations to foster effective communication and coordination [24]. The lack of a unified approach to MLOps and the absence of standardized practices can lead to misalignment among teams and hinder productivity [4]. Organizational culture, including team collaboration and shared understanding, plays a critical role in overcoming these challenges [25]. In environments where there are gaps in skills or collaboration, MLOps adoption is slowed down, and the lack of continuous learning and shared goals may result in the inefficiency of ML systems and processes. Furthermore, the complex and evolving nature of MLOps practices requires ongoing training and adjustment of workflows, which may not be prioritized without a culture of continuous improvement and cross-functional teamwork [20,26].

2. Lack of Expertise and Resources: organizations often struggle to find individuals with the required expertise to implement MLOps effectively, leading to significant challenges in execution. The interdisciplinary nature of MLOps demands a combination of skills in data science, software engineering, and operations, which can be difficult to source and coordinate within organizations [23,27]. In addition to the shortage of skilled professionals, the availability of necessary computing resources presents another challenge. The scaling of machine learning systems and the efficient management of large datasets require considerable infrastructure, which many organizations lack or cannot afford [1, 9]. Moreover, integrating multiple ML components across organizational boundaries introduces concerns around data privacy and governance [28]. The complexity of MLOps, coupled with the need for continuous monitoring, secure data sharing, and maintaining performance in multi-organization environments, further underscores the challenge of aligning technical resources with operational demands [22,29].
3. Difficulties in Handling and Managing: data efficient distribution, parallelization, and orchestration of data and tasks are crucial challenges in MLOps implementation. Managing large datasets in real-time environments, especially in complex machine learning workflows, requires robust data handling strategies. Organizations face difficulties in managing data within big data ecosystems, including tasks such as distributing and parallelizing data for processing [23]. These challenges are further complicated by the interdisciplinary nature of MLOps, where coordination across various teams with different expertise is required to streamline data management processes [6,25]. In addition, the integration of multiple tools and the need for continuous monitoring and orchestration of tasks can strain resources and impede the efficient flow of data [28]. Furthermore, ensuring scalability when dealing with vast amounts of data, while maintaining synchronization across different pipelines, adds an additional layer of complexity [8]. Data quality and governance issues, including the lack of access to clean and labeled data, are prevalent in MLOps systems, as noted by [30,31], further complicating data handling. Addressing these issues often requires a combination of automated solutions and advanced infrastructure management to optimize the flow and processing of data, which can be resource-intensive and technically demanding. The lack of standardized practices and tools for handling such complex data tasks [10], highlights the importance of developing effective MLOps strategies to meet the demands of real-time and large-scale data environments.
4. Difficulties in Automation and Workflows: implementing MLOps presents numerous challenges in automation and workflow management. One of the primary difficulties lies in the manual operations that hinder the efficiency of machine learning (ML) pipelines. ML projects often rely on manual processes for tasks such as testing, deployment, and retraining, making them prone to errors and inefficiencies [1,6]. Streamlining these operations through automation is crucial for reducing complexity and ensuring timely responses to iterative changes in data [10, 30]. However, achieving effective workflow automation requires overcoming hurdles in integrating different tools, libraries, and frameworks, which can create compatibility issues [29]. Moreover, managing the retraining of models is another significant challenge. Constant retraining is necessary as new data becomes available, but ensuring that models remain accurate and efficient requires continuous monitoring and handling of model artifacts [9,19]. Data versioning and model artifacts must be managed across various stages of the workflow, which becomes particularly difficult in environments with large datasets and multiple stakeholders [31,32]. Furthermore, scalability and infrastructure management become critical when automating workflows, especially in production environments where ML models need to operate seamlessly across diverse platforms [25,33]. The absence of established guidelines and standardized practices for MLOps further complicates the automation process [20].
5. Difficulties in Integration and Broader Scaling: integrating MLOps effectively into organizational processes and workflows poses several challenges, particularly in terms of scalability and fluctuating demands. Accurately predicting the infrastructure resources required for machine learning (ML) projects can be difficult, leading to scalability issues as data volumes and resource needs fluctuate [1]. Similarly, the lack of clear methodologies for MLOps, which contributes to difficulties in defining and implementing robust practices across organizations [16]. The complexity of integrating MLOps with existing processes is further compounded by the absence of standardized approaches which leads to inconsistencies and inefficiencies in implementation [17]. Furthermore, managing diverse tools and libraries, creates additional hurdles for teams trying to streamline their ML workflows [6,10]. These challenges are exacerbated when scaling ML systems, particularly in dynamic environments where infrastructure management becomes increasingly complex [8]. Integrating MLOps into multi-organization setups introduces data privacy and governance concerns that complicate cross-functional coordination [28].
6. Unexpected Performance and Thorough Monitoring: unexpected performance issues and the necessity of thorough monitoring present significant challenges in MLOps implementation. Frequent changes in data can disrupt model performance even when the collected data is error-free, emphasizing the need for robust monitoring practices [6]. Similarly, effective monitoring is crucial for detecting anomalies, ensuring compliance with regulatory requirements, and maintaining transparency throughout the ML life cycle [8]. Failures in production pipelines can stem from both ML infrastructure and software [19], making it essential to track model robustness and performance metrics accurately. Furthermore, continuous monitoring and maintenance is important to address challenges such as concept drift and data distribution shifts, which can significantly degrade model performance over time [29]. The integration of comprehensive monitoring practices is critical to adapting models to new data distributions and ensuring their reliability in dynamic production environments [31]. Thus, proactive and thorough monitoring not only mitigates the risks of unexpected performance issues but also ensures the sustainability and efficiency of MLOps workflows.
7. Poor Collaboration and Communication: this presents significant challenge in MLOps implementation. Communication issues between machine learning (ML) experts and the broader project team often result in silos where ML experts operate independently, detached from the overall project and customer needs [4]. This detachment hinders effective collaboration and slows the alignment of ML development with project goals. Similarly, gaps between domain experts and technical experts can impede the creation of meaningful datasets and may reinforce biases [27]. These gaps are further compounded by a lack of coordination where collaboration and communication between data scientists, ML engineers, and DevOps teams are essential but challenging to achieve [8]. The interdisciplinary nature of MLOps requires diverse professional roles and skills to work in sync, which can be hindered by undefined and evolving practices [34,35]. These collaborative challenges are further exacerbated by inconsistent communication during critical phases [33], which can lead to delays, misaligned priorities, and inefficiencies in MLOps workflows. Addressing these issues requires fostering a culture of collaboration, implementing clear methodologies, and improving communication channels to bridge gaps across teams.

Table 5
MLOps challenges in the literature.

Challenge		Description	References
C1	Lack of Necessary Skills and Culture	The need for a combination of cross-functional skills and a collaborative culture within organizations.	[23], [8], [19], [25], [20], [4], [22], [26], [24]
C2	Lack of Expertise and Resources	Organizations face difficulties in finding individuals with the required expertise. Additionally, challenges include obtaining the necessary computing resources and ensuring security and privacy measures are in place.	[23], [1], [28], [27], [9], [25], [34], [22], [24], [29]
C3	Difficulties in Handling and Managing Data	Challenges include efficient distribution, parallelization, and orchestration of both data and tasks to optimize the MLOps process.	[23], [1], [6], [10], [8], [30], [19], [28], [9], [25], [3], [11], [22], [38], [26], [24], [29], [39], [31], [32]
C4	Difficulties in Automation and Workflows	This includes manual operations that need to be streamlined, implementing effective workflow automation, managing the retraining of models, and handling artifacts throughout the process.	[1], [6], [10], [30], [19], [9], [25], [20], [3], [11], [29], [39], [33], [31], [32]
C5	Difficulties in Integration and Broader Scaling	Challenges encompass integrating MLOps effectively into various organizational processes and workflows, addressing scalability issues, and dealing with fluctuating demands. Additionally, there is a lack of clear methodologies, posing challenges in defining and implementing MLOps.	[1], [16], [6], [10], [8], [28], [40], [22], [26], [35], [17]
C6	Unexpected Performance and Thorough Monitoring	Challenges involve monitoring the production environment, addressing issues such as bias, mental health risks, and ensuring explainability in model outputs. Additionally, challenges include monitoring model performance, dealing with changes in data, and the lack of automation in testing and deployment.	[23], [6], [10], [8], [19], [28], [9], [25], [20], [40], [3], [4], [11], [22], [38], [29], [39], [32]
C7	Poor Collaboration and Communication	Challenges include fostering collaboration and communication between data scientists, ML engineers, and DevOps teams. Ensuring alignment, traceability, and reproducibility throughout the ML life cycle is emphasized.	[6], [10], [8], [19], [28], [27], [20], [4], [11], [22], [34], [35], [39]
C8	Handling Managerial Challenges	involve addressing gaps between domain and technical experts, slow problem-solving iterations, and effective management of infrastructure resources required for MLOps.	[6], [10], [8], [27], [37], [25], [40], [3], [4], [33], [29], [17]

8. Handling managerial challenges: this challenge requires addressing various organizational and operational barriers to ensure successful implementation in MLOps. A significant issue is the communication gap between domain experts and technical experts, as their differing knowledge and objectives can hinder collaboration and data quality, potentially reinforcing biases [4,27]. The interdisciplinary nature of MLOps adds complexity, as successful execution demands seamless collaboration among diverse professional roles and technical teams [35]. Additionally, organizations often face challenges related to organizational culture, skill gaps, and the lack of well-defined and evolving MLOps practices, which can delay problem-solving and hinder efficiency [24,27]. Slow iterations caused by ineffective architectures and manual data pipelines further impede agility in ML workflows [27]. Integrating MLOps practices into existing organizational processes also requires cultural shifts and significant change management efforts [8,36]. Moreover, maintaining synchronization between development and production teams is critical, as poor coordination can lead to delays and inefficiencies [6]. Finally, resistance to change and the rapid proliferation of tools and frameworks pose additional hurdles for managerial teams [36,37].

Table 5 outlines the previous challenges by breaking those challenges into eight categories followed by their description and the references pointing to the specific challenge.

4.2. RQ2- What are the practices in MLOps for developing, deploying, and maintaining machine learning models?

The identified key practices for successful MLOps implementation are summarized in Table 6. These practices encompass the entire machine learning life cycle, from building and training models to deployment and monitoring. The identified key practices include

1. Automating CI/CD: CI/CD automation is a fundamental aspect of MLOps that aims to streamline the development, deployment, and maintenance of machine learning (ML) models. It encompasses automated processes such as continuous integration (CI), continuous deployment (CD), and continuous training (CT) to facilitate seamless and efficient operations throughout the ML life cycle. CI/CD pipelines ensure that model changes, whether in code, data, or configurations, are continuously tested, validated, and deployed with minimal manual intervention, significantly reducing time-to-market and improving model reliability [1,6,8,10]. Automation of these processes enables faster iteration, allowing models to be quickly retrained and redeployed in response to new data, ensuring that the system remains up-to-date and accurate in real-world scenarios [6,19,41]. Additionally, containerization tools such as Docker and orchestration tools such as Kubernetes are widely used within these pipelines to improve model portability and scalability across different environments, making it easier to manage large-scale deployments [6,10]. Furthermore, CI/CD automation supports collaboration between development, operations, and other stakeholders, fostering smoother workflows and enabling rapid delivery cycles [10,11,42]. By incorporating these practices, MLOps ensures that ML systems can be maintained efficiently and continuously improved in production environments [1,9,42].
2. Orchestrating Workflows: Orchestrating workflows is a critical component of successful MLOps implementation, enabling the automation and management of end-to-end machine learning pipelines. Workflow orchestration ensures that data ingestion [25], model training, testing, deployment, and monitoring processes are seamlessly integrated and automated. This allows for more efficient collaboration among team members and accelerates the release of ML models. Key practices in workflow orchestration include the use of continuous integration (CI) and continuous deployment (CD) pipelines, which automate the testing, validation, and deployment of models [1,6,8,20]. Tools such as Docker and Kubernetes are used to package and scale ML applications in isolated environments, enhancing portability and reproducibility [6,40]. Furthermore, continuous training (CT) is crucial for retraining models when new data become available, ensuring that models stay up to date and performant [6,8,10]. Automation also reduces manual efforts and improves

the accuracy and reliability of deployments by incorporating testing, monitoring, and logging mechanisms throughout the ML life cycle [10,16,42]. In addition, incorporating feedback loops, versioning, and monitoring practices [1,22,41] ensures that models evolve in response to real-world data and operational requirements. As automation and orchestration become increasingly critical, MLOps frameworks such as TensorFlow Extended (TFX), Kubeflow, and AWS SageMaker provide the infrastructure necessary to manage complex ML workflows [32,43]. These tools support the efficient and scalable execution of machine learning tasks, further streamlining the deployment and operationalization of models.

3. Ensuring Reproducibility: Reproducibility is a critical aspect of ensuring the reliability and consistency of machine learning models and their deployment in operational environments. Containerization tools such as Docker play a vital role in creating reproducible environments, allowing experiments to be consistently replicated across various systems [6]. Additionally, version control mechanisms for code, data, and configurations are essential to maintain reproducibility throughout the machine learning life cycle [9]. Automated tracking and model management systems further enhance reproducibility by ensuring that different model training runs are executed under consistent conditions [44]. Standardizing machine learning pipelines through automation helps mitigate discrepancies and ensures that results are replicable across different stages of the model development process [20]. Furthermore, tools such as GitHub Actions and Continuous Machine Learning (CML) enable the automation of workflows, contributing to reproducibility [42]. Finally, automated reporting and logging systems ensure that all parameters, configurations, and outcomes are comprehensively documented, fostering transparency and facilitating the reproducibility of experiments [45].
4. Versioning: Versioning is a key practice in MLOps, ensuring that machine learning models, data, and configurations are consistently tracked and managed throughout their life cycle. [9,10,20,22] highlight the importance of versioning in maintaining the reproducibility and repeatability of machine learning workflows. By versioning models, data, and code, organizations can easily track changes, roll back to previous versions, and ensure that experiments can be replicated. This practice is vital for continuous integration and deployment (CI/CD) pipelines, as it allows efficient management of model updates and data transformations [6,25,42]. Furthermore, [46] emphasizes that version control is critical for managing data and model life cycles, ensuring that the latest versions are always available for testing and deployment, while [32,43] identify tools such as DVC and MLflow, which facilitate version control for data and models in MLOps pipelines. These tools allow data scientists and engineers to track datasets, model parameters, and experiment configurations, fostering collaboration, enhancing transparency, and ensuring that models can be reliably deployed at scale. Through proper versioning, organizations can improve model stability and traceability, thereby enhancing the overall quality and performance of machine learning systems deployed.
5. Collaboration: Collaboration plays a pivotal role in the successful implementation of MLOps, as it bridges the gap between different stakeholders and ensures smooth integration of processes. Effective communication between business and technology experts, as well as within development and deployment teams, is essential to align business goals with technical solutions [1,27]. Using no-code AI platforms, domain experts can participate directly in the ML workflow, promoting collaboration without the need for deep programming expertise [27]. Furthermore, fostering collaboration through automation tools, such as GitHub Actions [42], and incorporating monitoring and logging [10], ensures continuous testing and validation of models, leading to reliable and scalable solutions. MLOps emphasizes the importance of continuous integration (CI) and continuous deployment (CD), which streamline collaboration by automating repetitive tasks, thus reducing human errors and enabling faster delivery cycles [4,10]. Collaboration is also facilitated by using open source frameworks and DevOps tools, which allow seamless integration of ML pipelines across diverse environments and support continuous training and monitoring of the models [6,20,45]. By combining these practices, organizations can ensure that the collaboration between various stakeholders is optimized, driving the continuous improvement of ML models and solutions throughout their life cycle [16,40,43].
6. Continuous Training and Evaluation:Continuous training and evaluation are vital components of successful MLOps frameworks, ensuring that machine learning models remain accurate and performant in real-world environments. Continuous training (CT) involves automating models retraining based on new data or changing conditions, which is crucial to maintaining model relevance and performance [6,8,16]. Using CI/CD pipelines in automating CT, using tools such as Docker and Kubernetes to facilitate model deployment and scalability is crucial [6]. Furthermore, continuous evaluation is essential to monitor model performance over time, detect performance degradation, and ensure that models remain robust and accurate in production [20]. Continuous monitoring, testing, and feedback loops, key elements of CT and continuous evaluation, are essential to ensure long-term success of the ML models deployed [9,10,40]. Effective data management, versioning, and collaboration are also vital in sustaining continuous training and evaluation processes [42,43]. Through the integration of these practices, organizations can create adaptive, efficient, and reliable ML systems that continually improve based on real-world feedback.
7. Track and Log ML Metadata: this practice enables effective management and accountability throughout the ML life cycle. Metadata includes information about datasets, models, configurations, hyper-parameters, and performance metrics, which helps ensure reproducibility, traceability, and transparency of ML workflows. Automating metadata management within CI/CD pipelines enables seamless integration, deployment, and monitoring, while improving model performance and reproducibility [6,10,40]. Additionally, continuous monitoring and logging are essential for capturing performance changes over time, providing information on model behavior in real-world environments [16,25]. Version control and automated testing are integral components of metadata tracking, ensuring that models can be efficiently reproduced and refined [9,22]. ML pipeline stages, including data collection, feature engineering, and deployment, critically rely on metadata tracking to maintain data quality and consistency [37]. The use of tools such as GitHub Actions, DVC, and MLflow offer robust solutions for tracking and logging metadata across different stages of the ML pipeline, from model design to deployment and maintenance [32,43].
8. Continuous Monitoring: CM is a pivotal component of MLOps frameworks, ensuring the reliability and performance of machine learning models in production environments. Continuous monitoring plays an important role in tracking model behavior, detecting performance degradations, and validating predictions over time [3,10,20]. Continuous monitoring supports maintaining the operational quality of deployed models by providing real-time feedback loops to address issues such as data drift and evolving production requirements [25,40]. continuous monitoring can be applied in regulated environments, such as the Oravizio system for clinical risk assessments, where monitoring ensured compliance with post-market surveillance standards [24]. The integration of open-source tools and automated

Table 6
Practices and the appointed articles.

Practice		Description	References
P1	Automating CI/CD	Continuous integration, delivery, and deployment are allowed by CI/CD automation. Build, test, deliver, and deploy are all completed by it. It allows developers to quickly receive feedback on whether a step was successful or unsuccessful, which boosts productivity overall.	[1], [16], [6], [10], [8], [19], [27], [9], [42], [20], [45], [34], [47], [40], [3], [4], [11], [41], [22], [26], [48],[29], [49], [39], [33], [37], [32], [43]
P2	Orchestrating workflows	An ML workflow pipeline's tasks are coordinated by workflow orchestration.	[1], [6], [8], [27], [25], [34], [41], [22], [46], [32], [43]
P3	Ensuring reproducibility	To be able to repeat an ML experiment and get the exact same findings is known as reproducibility.	[1], [9], [41], [29], [33], [46]
P4	Versioning	To provide for both reproducibility and traceability (for compliance and auditing purposes), versioning makes sure that data, models, and code are versioned.	[1], [10], [8], [19], [9], [42], [22]
P5	Ensuring collaboration and communication	Cooperation makes it possible to work together on models, code, and data. In addition to the technical side, this practice focuses on creating a collaborative and communicative work environment with the goal of reducing domain barriers across various positions.	[1], [10], [27], [40], [3], [4], [22], [35], [39]
P6	Continuously train and evaluate ML models	Periodically retraining the machine learning model using new feature data is known as continuous training, or CT. The assistance of an automated ML workflow pipeline, a feedback loop, and a monitoring component allows for continuous training. An evaluation run is always included in continuous training to evaluate the change in model quality.	[1], [16], [8], [19], [27], [9], [42], [25], [20], [45], [34], [47], [40], [11], [50], [32]
P7	Track and log ML metadata	Every coordinated ML workflow task has its metadata tracked and logged. For each training job iteration, metadata tracking and logging (e.g., training date and time, duration, etc.) is necessary to ensure the complete traceability of experiment runs. This includes model-specific metadata, such as used parameters and the resulting performance metrics, model lineage: data and code used.	[1], [10], [19], [42], [22], [29]
P8	Continuous monitoring	Continuous monitoring involves routinely evaluating the infrastructure, data, model, code, and model serving performance (such as prediction accuracy) in order to identify any flaws or modifications that may have an impact on the final product's quality.	[1], [16], [10], [19], [27], [9], [20], [4], [11], [41], [22], [24], [48], [32], [43]
P9	Implementing feedback loops	Integrating lessons from the quality assessment step into the engineering or development process necessitates many feedback loops. In order to facilitate retraining, an additional feedback loop between the scheduler and the monitoring component is needed.	[1], [9]

pipelines to facilitate seamless monitoring of deployed systems helps enhancing reproducibility, scalability, and collaboration across teams [22,45].

9. Implementing Feedback Loops: Incorporating feedback loops to continuously improve the ML development process, including retraining models and incorporating lessons learned from quality assessments [8]. This involves gathering feedback from various stakeholders, such as end-users and business stakeholders, and using this feedback to refine the ML models and the overall MLOps process.

4.3. RQ3 - What maturity models or frameworks have been proposed for assessing an organization's level of MLOps maturity, and how are they structured?

This inquiry delves into the existing frameworks designed to evaluate an organization's maturity in adopting MLOps practices. Despite receiving less attention in existing literature, this question holds significant importance as it offers insights into the structured approaches for assessing MLOps maturity, and it also gives an indication about importance of automation.

The maturity levels represent a progression of MLOps maturity within an organization. Each level signifies increasing sophistication in how an organization implements and manages its machine learning (ML) workflows. Note that various models define various maturity levels, which means not all models have 4 levels, a general description of these levels is as follows:.

1. Level 0: This is the baseline where there are no formal MLOps practices in place. ML model development and deployment are typically manual and ad-hoc, with limited automation or standardization.
2. Level 1: At this level, some basic MLOps practices are being adopted. This might include the initial automation of some parts of the ML pipeline, such as data preparation or model training. However, there is still significant manual intervention and a lack of overall coordination and standardization across the ML life cycle.
3. Level 2: This level indicates a more mature approach in which key MLOps practices are being implemented. This could include the use of CI/CD pipelines for automated model building and deployment, as well as basic monitoring of model performance. However, there may still be room for improvement in terms of overall integration and optimization of the MLOps processes.
4. Level 3: At this level, organizations have well-defined and automated MLOps processes in place. This includes robust CI/CD pipelines, automated model deployment, and comprehensive monitoring and feedback loops. The focus is on continuous improvement and optimization of the ML life cycle.
5. Level 4: This represents the highest level of MLOps maturity. Organizations at this level have fully automated and optimized MLOps processes, with a strong focus on continuous improvement and innovation. They leverage advanced technologies and methodologies, such as MLOps platforms and AI/ML platforms, to streamline their ML workflows and achieve business objectives.

For instance, Garg et al. [6] proposed three different levels of MLOps maturity, MLOps Level 0: This is the fundamental level of maturity, where workflows are manually script-driven at every stage of the machine learning life cycle. There is no automation, resulting in a lack of active performance monitoring. This level is suitable for simple ML solutions that do not require frequent updates. MLOps Level 1: At this

Table 7
MLOps maturity models' levels.

Maturity model	Level 0	Level 1	Level 2	Level 3	Level 4
Google model	Manual process	ML pipeline automation	CI/CD pipeline automation	NA	NA
Microsoft model	No MLOps	DevOps but no MLOps	Automated Training	Automated Model Deployment	Full MLOps Automated Operations
GigaOm Model	No DevOps practices adopted	DevOps practices such as CI/CD have been adopted for non-ML components	Development iterative but CI/CD not in place for models	DevOps practices such as CI/CD, code reviews in place for ML code	Comprehensive MLOps pipeline supporting frequent model updates
[51] model	Automated data collection	Automated model deployment	Semi-automated model monitoring	Fully-automated Model Monitoring	NA
[52] model	Initial	Repeatable	Defined	Managed	Optimization

level, the steps of machine learning experiments are orchestrated to automate the ML pipeline solely to undertake continuous model training. This level enables continuous model prediction services and automates retraining models in production using new data. However, the need for CI/CD solutions to automate the creation, testing, and deployment of ML pipelines is critical. MLOps Level 2: This level includes automation of a CI/CD pipeline for more rapid and reliable updating of pipelines in production, which requires a more robust automated system. This level achieves the most efficient and automated development workflow as it leverages industry-standard tools and techniques that have successfully delivered enterprise-level software solutions in the past decade. Similarly, John et al. [51] developed maturity model of 3 levels too.

Google model consists of three levels while Microsoft MLOps maturity level has 5 levels [19,32]. Note that the various MLOps maturity models are usually developed based on CMM and CMMI models as in [3,22,52–54]

Table 7 describes the levels of MLOps maturity models discussed before. It is worth mentioning that GigaOm maturity model unpacks each maturity level through 5 categories: strategy, architecture, modeling, processes, and governance. However, for the sake of clarity and ease, only the strategy category was explained.

4.4. RQ4 - What are the key lessons learned from successful and unsuccessful MLOps implementations?

This inquiry aims to extract valuable insights from past experiences to guide future MLOps efforts.

1. Lessons learned from successful MLOps implementation: successful implementations of MLOps emphasize a combination of cross-functional skills and collaboration among business stakeholders, solution architects, data scientists, data engineers, software engineers, DevOps engineers, and MLOps engineers [1,23]. Core principles include automation of ML pipelines, continuous model training, and automated retraining, which contribute to efficient deployment and monitoring [6,28,41]. Clear processes and guidelines for incorporating ML pipelines into software development, alongside scalability, high availability, flexibility, and maintainability, are critical [50,51]. Additionally, the use of CI/CD solutions ensures rigorous testing and smooth deployment of ML components, while robust governance frameworks ensure reliability and compliance [6,51]. Automated MLOps pipelines can achieve comparable accuracy to manual processes while significantly reducing time-to-market. This is facilitated by tools like Jenkins that accelerate model training and platforms like AWS SageMaker, Google Cloud, and Kubeflow that streamline end-to-end machine learning workflows [10,11,34]. Furthermore, monitoring and tracking of models in production, managing data distribution shifts, and improving data quality through cleaning and adaptation are essential for maintaining model performance [19,31]. Collaboration, standardization, and leveraging open-source tools play pivotal roles in enhancing productivity and consistency across implementations [3,36]. Organizations benefit from adopting reference architectures and integrating orchestration platforms to combine services efficiently while promoting scalability and automation [34,37].
2. Lessons learned from unsuccessful MLOps implementation: Unsuccessful MLOps implementations often stem from a combination of technical, organizational, and cultural challenges. Unrealistic expectations about the capabilities of data science, along with issues related to business alignment, architectural gaps, and process inefficiencies, can significantly hinder adoption in traditional industries [23]. Poor testing, deployment, and monitoring processes further exacerbate failures [6,41]. A lack of standardization and quality in model development and deployment practices frequently leads to unreliable outcomes [31,51]. Challenges such as handling new data for retraining [19], managing changes and versioning of multiple ML models simultaneously [35], and addressing accuracy degradation during online deployment due to incomplete or erroneous data are also critical barriers [48]. Additionally, over-reliance on reusing the same test cases can increase the risk of overfitting [31], while the absence of defined evolving practices and clear processes disrupts the integration and maintenance of ML pipelines [50]. Moreover, gaps in collaboration between cross-functional teams, inadequate governance frameworks, and limited automation often result in inefficient MLOps implementations [3,36].

Table 8 summarizes the identified lessons learned from both successful and unsuccessful MLOps implementations.

Table A.10 in the Appendix shows the list of references and the research questions answered by reference. X informs that the answer to the specific question was found in the paper.

4.5. Validity threats

4.5.1. Internal validity

Internal validity concerns the extent to which the study accurately measures what it intends to measure. In this study, potential internal validity threats include selection bias. The focus on successful MLOps implementations may have biased the findings towards a more positive view of MLOps practices and challenges, potentially overlooking common pitfalls and challenges faced in real-world scenarios. Furthermore, the findings may not be fully generalizable due to the potential influence of contextual factors such as organizational size, industry, project scope, and resource availability on MLOps practices and challenges.

Table 8
Key lessons learned.

Ref.	Lessons learned	
	Successful implementation	Unsuccessful implementation
[23]	– Cross-functional skills – Collaborative culture – Competent expertise	– Unrealistic expectations in the field of data science, and business – Business, architectural, process, and organizational challenges
[1]	– Distribution of roles	NA
[16]	– Choosing hyper-parameters – Model monitoring	NA
[6]	– Automation of ML pipelines – Continuous model training and automated retraining – CI/CD solutions are critical for testing and releasing new ML components	– Lack of proper testing and deployment processes
[10]	– Usage of automated MLOps pipelines – Usage of Jenkins parallel function	NA
[51]	– Collaboration between different roles – Applying architectural principles	– Lack of standardized and quality model development
[19]	– Model monitoring and tracking	– Struggle with data manipulation for retraining
[20]	– Rapid and frequent deployment of ML models – MLOps reduces the time to develop a service – MLOps ensures a safe transition from model development to deployment – Collaboration and teamwork	NA
[34]	– MLOps tools support automation and continuous delivery of new models – AWS Sagemaker, Azure ML, and Dotscience have a high maturity level in ML process – Manual steps are still needed for model compatibility and issue resolution – Gradually implement each step before aiming for full automation – Evaluation in production is necessary to fully analyze MLOps tool feature – Orchestration platforms allow for combining different services	NA
[40]	– Continuous monitoring and validation of AI systems	– Great amount of supporting infrastructure was needed – Difficulty in monitoring prediction quality – Looking at the distribution of predictions can help detect deviations.
[3]	– Open-source tools are more modular and often offer higher quality – Benchmarking open-source tools can help in selecting problem-specific solutions	– Commercial platforms may not provide a flexible and transparent workflow – Multiple iterations and spending money on ineffective solutions can be avoided – No single tool can realize a fully automated MLOps workflow.
[11]	– DataRobot MLOps platform offer great functionalities – MLOps on Google Cloud Service offers decent service – Kubeflow pipelines automate ML deployments	NA
[41]	– A fully automated pipeline for continuous training – Competent ML professionals – Identify MLOps challenges by surveys and data analysis	– Lack of proper testing, deployment, and monitoring components
[50]	– Clear processes and guidelines – Long-term maintainability and evolvability – Standardized tools and frameworks for MLOps	NA
[38]	– Continuous training and serving – Importance of a monitoring solution – Importance of automation – Orchestrated experiment environment for model behavior validation.	NA
[24]	– Continuous software development approaches – Automation and monitoring – Post-mortem analysis	– Regulatory constraints
[35]	– Versioning and managing changes	– Lack of well-defined practices – Differing expectations and perspectives in software system
[48]	– Handling erroneous or incomplete data and addressing accuracy degradation – Prioritize automation, quality monitoring, and retraining challenges	NA

(*continued on next page*)

Table 8 (*continued*).

[54]	– Use of anti-patterns – Understanding broader context in ML	NA
[31]	– Importance of data quality – Improving the quality and quantity of training data – Adopting CI	– Neglecting over fitting
[37]	– Consider a reference architecture and a common set of requirements	NA
[36]	– Containerization and version control systems – Dedicated MLOps teams and DevOps integration – Cultural transformation – Utilizing MLOps frameworks and open-source tools	NA

Table A.9
Result of the evaluation of quality criteria.

Study reference	Quality criteria					Total
	1	2	3	4	5	
[23]	1	1	0	1	0	3
[1]	1	1	0	1	1	4
[16]	1	1	0	1	1	4
[6]	0	0	1	1	0	2
[10]	0	1	0	1	1	3
[8]	0	1	0	1	1	3
[30]	0	0	0	1	1	2
[51]	1	1	0	1	1	4
[19]	0	0	1	1	1	3
[28]	1	1	0	1	1	4
[27]	0	1	0	1	1	3
[9]	0	0	0	1	1	2
[52]	1	1	0	1	1	4
[42]	0	0	0	1	1	2
[25]	1	1	0	1	1	4
[20]	0	1	0	1	1	3
[45]	1	1	0	1	1	4
[34]	0	1	0	1	1	3
[47]	1	1	0	1	1	4
[40]	1	1	0	1	1	4
[3]	1	1	0	1	1	4
[4]	0	1	0	1	1	3
[11]	1	1	0	1	1	4
[41]	1	1	0	1	1	4
[22]	0	1	1	1	1	4
[53]	1	1	0	1	1	4
[50]	1	1	0	1	1	4
[38]	0	1	0	1	1	3
[26]	1	1	0	1	1	4
[24]	1	1	0	1	1	4
[35]	1	1	0	1	1	4
[48]	1	1	0	1	1	4
[29]	1	1	0	1	0	3
[49]	0	1	0	1	0	2
[55]	1	1	0	1	0	3
[54]	0	1	0	1	1	3
[39]	1	1	0	1	1	4
[33]	1	1	0	1	1	4
[31]	1	1	0	1	1	4
[17]	0	1	0	1	0	2
[46]	1	1	0	1	1	4
[37]	1	1	0	1	1	4
[36]	1	0	0	1	1	3
[32]	1	0	1	1	1	4
[43]	1	1	0	1	1	4

Table A.10
Pre-selected studies.

Reference	RQ1	RQ2	RQ3	RQ4
[23]	X			X
[1]	X	X		X
[16]	X	X		X
[6]	X	X	X	X
[10]	X	X		X
[8]	X	X		
[30]	X		X	
[51]			X	X
[19]	X	X	X	X
[28]	X			
[27]	X	X		
[9]			X	X
[52]			X	
[42]		X		
[25]	X	X		
[20]	X	X		X
[45]		X		
[34]	X	X		X
[47]		X		
[40]	X	X		X
[3]	X	X	X	X
[4]	X	X		
[11]	X	X		X
[41]		X		X
[22]	X	X	X	
[53]			X	
[50]		X		X
[38]	X			X
[26]	X	X		
[24]	X	X		X
[35]	X	X		X
[48]		X		X
[29]	X	X		
[49]		X		
[55]		X		
[54]	X		X	X
[39]	X	X		
[33]	X	X		
[31]	X			X
[17]	X			
[46]		X		
[37]	X	X		X
[36]	X			X
[32]	X	X	X	
[43]		X		

4.5.2. External validity

External validity refers to the generalizability of the study findings to other contexts and populations. Limitations to external validity in this study include the potential for limited sample size. The number of included studies may not be sufficiently large or diverse to capture the full range of MLOps practices, challenges, and maturity models across different industries and project types. Publication bias, where studies with positive or successful outcomes are more likely to be published, may have skewed the findings. Additionally, the focus on recent literature (2019–Jan. 2024) may not fully capture the evolving nature of MLOps and the emergence of newer practices and challenges.

4.5.3. Construct validity

Construct validity concerns the extent to which the study measures the intended constructs. In this study, limitations to construct validity include the potential for mono-method bias. Relying solely on peer-reviewed academic literature may have limited the breadth and depth of the findings. Incorporating data from other sources, such as industry reports or case studies, could have provided a more comprehensive understanding of MLOps practices. Focusing solely on peer-reviewed

academic articles may have excluded valuable insights from industry blogs, white articles, and other relevant sources, potentially leading to an incomplete and potentially biased view of MLOps practices and challenges.

5. Conclusion

This research aimed to shed light on the best practices, challenges, and maturity models within the dynamic field of MLOps. By analyzing 45 peer-reviewed articles, we identified nine key MLOps practices, including automation, collaboration, continuous monitoring, and data management. Furthermore, we categorized and analyzed numerous challenges encountered in MLOps implementation. The study also explored prominent maturity models, such as those proposed by Google and Microsoft, providing insights into the evolving landscape of MLOps maturity.

While this research provides a comprehensive overview of current MLOps practices and challenges, it is important to acknowledge certain limitations. The focus on peer-reviewed literature may have introduced publication bias, as studies with positive or successful outcomes are more likely to be published. Additionally, the findings may not be fully generalizable due to potential contextual factors and the limited scope of the included studies.

Future research directions include applying MLOps practices to other domains such as Computer Vision Operations (CVOps), data management, cloud continuum, and networking. Mapping best practices to maturity models and investigating the impact of MLOps maturity on organizational performance are also areas for further exploration. Furthermore, research on the ethical considerations of MLOps, including data privacy, bias, and fairness, is crucial for responsible and ethical AI development and deployment.

This research contributes to the growing body of knowledge surrounding MLOps by providing valuable insights into current practices, challenges, and maturity models. The findings of this study can serve as a valuable resource for practitioners, researchers, and organizations seeking to implement and improve their MLOps capabilities.

CRediT authorship contribution statement

Mohammad Zarour: Supervision, Data curation. **Hamza Alzabut:** Writing – review & editing, Writing – original draft, Formal analysis. **Khalid T. Al-Sarayreh:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Tables A.9 and A.10

See Tables A.9 and A.10.

Data availability

No data was used for the research described in the article.

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